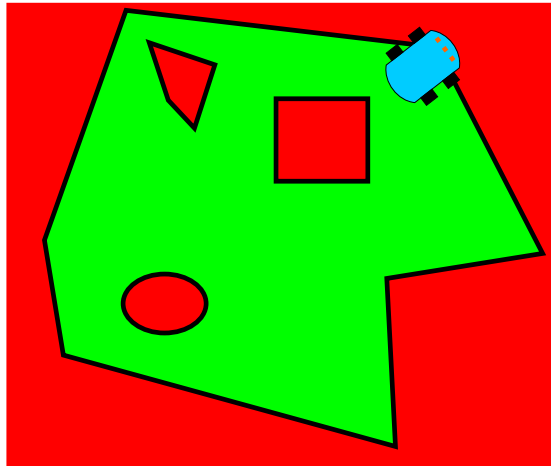


CCPS 607 – Autonomous Mobile Robotics

Lab 1 – Stay on the surface

The Environment and Concepts:

There exists a flat world of irregular shape. The world is surrounded by a void and if your robot falls into the void, it is gone forever. Somewhere on this world is a puck, a small, light object that needs to be disposed of. In the reality of the lab, there will be a surface raised to some height and if your robot falls off, the TA may not let you catch your robot. In the example figure below, black shows the edge of the surface, green is the area that the robot can be on and red represents void. Your robot might have part of its body in the void, but you obviously don't want it to fall off. This is only an example; the test environment may be somewhat different.



Requirements:

- Working in groups of one or two, you are to build a robot capable of surviving in the world while in (near) continuous motion for a period of no less than 2 minutes without falling off the “edge.”
 - The formal definition of near continuous motion is that the robot should always be moving with any pauses in movement lasting no longer than 0.5 seconds.
 - Your robot must, at some point during the demo, push the puck off the edge. No specific detection of the puck is required – random movement will be enough to accomplish this task.
- Aside from demoing your robot, you must make a short video (less than 2 minutes) of your robot on the specified environment.
- You must create a writeup (1 page maximum) describing of your algorithm/strategy and cover these points: Describe the overall design of the robot. Describe what sensors you used, why they were chosen, and the placement of the sensors and why that is important.

Submissions:

- You must demo your robot at the arranged time slot.
- You must submit your video on D2L on the day of demo.
- You must submit the 1-page writeup on D2L on the day of demo.

Late submission is possible. Since this has to be demoed, the only other time to submit would be the following week. You'll get a 25% deduction for late submission. Anything later than 1 week will not be accepted.

Rubrics:

The lab will be marked out of 40. All marks will come from the demo, but no marks will be given without submission of the writeup.